On a mixed GPU cluster, put the GPU model in every Slurm request

For HPC and ML platform teams running Slurm

Step 1: find the GPU type names your cluster already uses

01

List GPUs per node

sinfo -o "%N %G" prints each node's generic resources, with the GPU type where the admin set one.

02

Copy the admin's names

Types are set in gres.conf. For auto-detected GPUs, a type must match all or part of the name slurmd reports.

Step 2: ask for a type and a count in one request

01 **--gres=gpu:h100:4**

Slurm's form is name[:type:count]. A bare --gres=gpu:4 states a count and no model.

02 **--gpus=h100:4**

The --gpus options take the same [type:]count form. The sbatch docs use --gpus=volta:3.

Step 3: keep every job step on the type the job asked for

01

Match srun to sbatch

Slurm wants typed and untyped GPU requests to be consistent within a job. Use the same typed form in sbatch and in every srun step.

Step 4: give each part of a workflow its own GPU type in a heterogeneous job

01

One job, several components

Split components with a colon on the sbatch line or a #SBATCH hetjob line. Each sets its own type and count.

02

Know the limits

Heterogeneous jobs cannot be job arrays, and only the backfill scheduler places them.

What we would tell a team starting today

> **Put the GPU type in the job script, and the GPU name in the job log.**
>
> Our speech fine-tune had to request its GPU by GRES on the shared cluster it ran on. A typed request keeps the GPU model in the job script, next to the result.

How do you pick GPU types in your Slurm scripts today?

Scoping is short, fixed and ends in a written plan.

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